Machine Learning Methods to Predict Gaming Addiction Risk Level

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Abstract— In this paper, a dataset on the various day-to-day and behavioral features was used to predict risk level for video game addiction, or Internet Gaming Disorder (IGD). A subset of features was selected from the dataset to match symptoms commonly associated with IGD, and Exploratory Data Analysis and machine learning methods were applied to the dataset to determine how viable it was for classification. Results indicated that meaningful groupings were present in the dataset, and that the symptom subset could be used to accurately predict addiction risk level using a 2-layer Multi-Layer Perceptron model to 95.5% accuracy.

Keywords—machine learning, Exploratory Data Analysis (EDA), logistic regression, k-means clustering, Multi-Layer Perceptron (MLP)

GitHub Link: https://github.com/dianajohnson49/ml-project/

1. INTRODUCTION

As access to the internet and computer technology has grown easier, new behavioral disorders such as video game addiction have grown more prevalent. A paper published on the National Library of Medicine's website associated video game addiction with Internet Gaming Disorder and noted that it can cause changes in the brain and disruptions to day-to-day behavior in young people [1].

This paper, titled "Symptoms, Mechanisms, and Treatments of Video Game Addiction," identified several changes in habitual routines present in those with a video game addiction, such as loss of sleep, social isolation, and lower grades in an academic setting [1]. For this project, a dataset was identified which included features for some of the same symptoms mentioned in [1], where the prediction target was "gaming addiction risk level" [2]. The first task for this project was to evaluate the dataset, titled "Gaming and Mental Health," to determine whether it would be viable for prediction. The second task was to determine whether there was correlation between features in the dataset corresponding to those mentioned in the National Library of Medicine paper, and whether these features could be used to accurately predict the gaming addiction risk level for an individual.

Machine learning and exploratory data analysis methods such as K-means clustering, Principal Components Analysis (PCA), Spearman correlation, Logistic Regression models, and Multi-Layer Perception models were used to accomplish these tasks.

2. DATASETS AND EDA

Two datasets were initially identified: one titled "Mental Health Disorder Risk Assessment Dataset" [3] and another titled "Gaming and Mental Health" [2]. The original project goal was to use the "Mental Health Disorder Risk Assessment" and identify features similar to those mentioned in [1], but EDA performed on this dataset indicated that it was not suitable for classification. The EDA performed on the "Gaming and Mental Health" dataset had promising results, so this dataset was the one selected for use on this project.

2.1 Mental Health Disorder Risk Assessment Dataset

The initial dataset was a synthetically generated mental health risk assessment dataset based on 25 features including mental health risk, the target class. Some of the features included anxiety score, age, financial stress level, and social support score. There were 9 categorical and 16 numerical features. The data contained approximately 25,000 samples, and no missing values. It was decided that this dataset was unfit for classification after performing EDA (See Section 2.2).

2.1.1 Preprocessing and Exploratory Data Analysis

Both clustering and Orange were used as EDA tools on the original dataset from [3]. In Orange, the different feature correlations, distributions, and statistics were calculated

and visualized. After calculating the spearman correlations for all features, the results were all near 0. The highest correlation was between anxiety score and mental health risk at 0.32. The distributions of each feature using the target were also evaluated (see Figure 2.1). As shown, the various features were evenly distributed across the target classes, which indicated that the dataset would be unfit for a prediction task. The attempt at clustering the data was also indicative of a poor dataset.

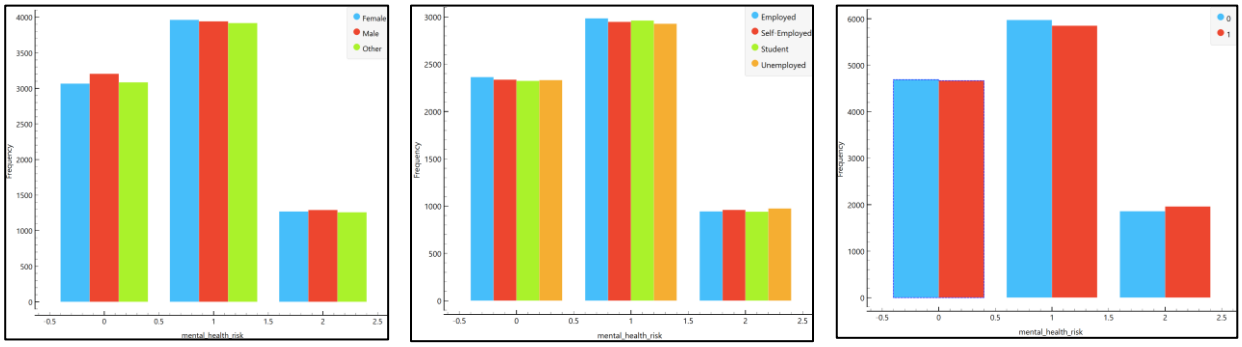

Figure 2.1: A comparison of different feature distributions per level of mental health risk (*left: gender, middle: employment status, right: substance use)*

2.2 Gaming and Mental Health Dataset

Since the features in the original dataset were unfit for classification, a new dataset from [2] focused on gaming addiction risk was used. The dataset had 27 features including the target variable, gaming_addiction_risk_level. The target included four risk levels: low (0), moderate (1), high (2), and severe (3). Of the 26 non-target features, 5 were categorical and the other 21 were numeric.

2.2.1 Preprocessing and Exploratory Data Analysis

The preprocessing steps included imputation, conversion of categorical data to numeric, and the removal of the ID feature (record_id). The record_id was removed before performing EDA and each categorical variable was converted to numeric using the Numpy and Pandas libraries in Python. 572 values were missing from the features grades_gpa and work_productivity_score, so values were imputed for those missing using either random generation or the mode.

The class distribution for the four levels of risk is shown in Figure 2.2. The low-risk class had around 500 samples, as compared to the severe risk class which had around 150. This shows that there was a significant class imbalance between the low-risk subjects compared to the other moderate- to severe-risk subjects.

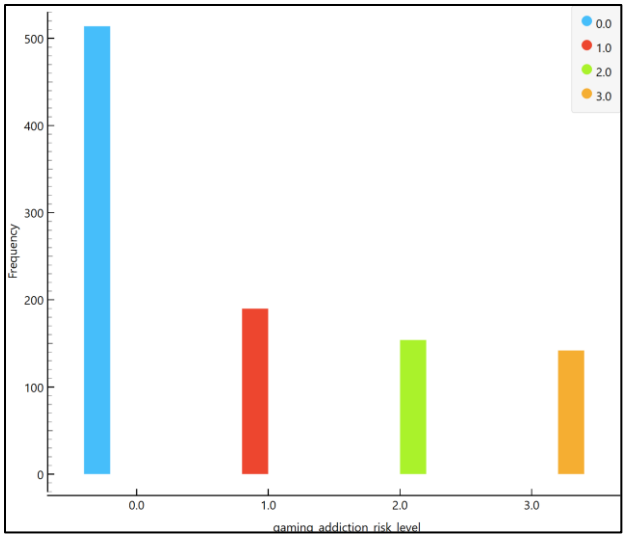

Figure 2.2: Distribution of different classes in the dataset

EDA was performed using Orange and Python. Spearman correlations were calculated in Orange and revealed high correlations between several features and the target, as seen in Table 2.1. The features most highly correlated with the target were Daily Gaming Hours, Social Isolation, Social Hours Weekly, Sleep Hours, and Weekly Exercise. This coincides with the results from [1]. Academic Performance was also included since it corresponded to a symptom identified in [1], even though it did not have a correlation as high as the other features used. Table 2.2 shows the full spearman correlations for the subset. This 6-feature subset was used for prediction later in the project to support the conclusions drawn in [1].

Feature Name	Reference Name	Correlation to Target
daily_gaming_hours	Daily Gaming Hours	0.857
social_isolation_score	Social Isolation	0.757
face_to_face_social_hours_weekly	Social Hours Weekly	-0.746
sleep_hours	Sleep Hours	-0.667
exercise_hours_weekly	Weekly Exercise	-0.645
Academic_work_performance	Academic Performance	-0.483

Table 2.1: Correlations to target variable

	Social Isolation	Social Hours Weekly	Sleep Hours	Weekly Exercise	Daily Gaming Hours	Academic Performance	Gaming Addiction Risk
Social Isolation	1	-0.776	-0.682	-0.643	0.866	-0.492	0.757
Social Hours Weekly	-0.776	1	0.693	0.642	-0.872	0.488	-0.746
Sleep Hours	-0.682	0.693	1	0.55	-0.778	0.487	-0.667
Weekly Exercise	-0.643	0.642	0.55	1	-0.734	0.403	-0.645
Daily Gaming Hours	0.866	-0.872	-0.778	-0.734	1	-0.551	0.857
Academic Performance	-0.492	0.488	0.487	0.403	-0.551	1	-0.483
Gaming Addiction Risk	0.757	-0.746	-0.667	-0.645	0.857	-0.483	1

Table 2.2: Spearman correlations for 6-feature subset

Figure 2.3 shows the distribution of the target class across the selected 6-feature subset. By visual inspection, there appear to be distinct relationships between the different features and the target classes. For example, the low-risk class is associated with the lower range of Daily Gaming Hours, whereas the moderate- to severe- risk classes are associated with higher daily gaming hours.

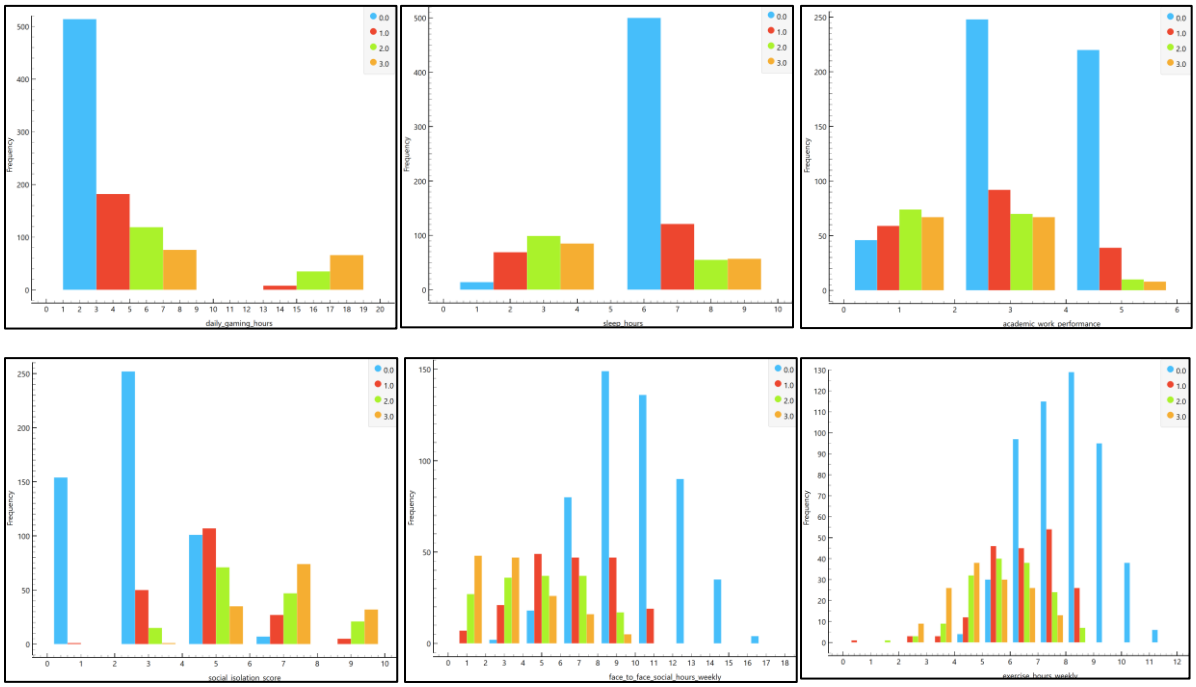

Figure 2.3: Histograms showing the target for different features from Table 2.1

The final dataset exported from orange included the removal of the record_id, imputation of missing values, and conversion of categorical data to numeric. This final dataset was used for all consecutive tasks.

Because there were 25 non-unique features in the dataset, the use of a dimensionality reduction method was necessary before the dataset could be visualized. Of the possible dimensionality reduction methods, t-distributed Stochastic Neighbor Embedding (t-SNE) was chosen for its ability to represent local structure and preserve clustering. Rather than just using distances to represent similarity, t-SNE calculates a conditional probability of similarity, or the probability that point X would choose point Y as its neighbor based on a Gaussian curve [4]. The dataset is then mapped to a lower-dimensional space, and the probability of similarity is recalculated. The placement of each datapoint in the lower dimensional space is optimized by reducing the difference between the high-dimensional probability of similarity and the low-dimensional probability of similarity [4].

To implement this method, the Scikit-learn and Pandas libraries were used in Python. The 25-dimensional dataset was reduced to a 3-dimensional representation using t-

SNE, and the data vectors were plotted using the Matplotlib library and can be seen below in Figure 2.4.

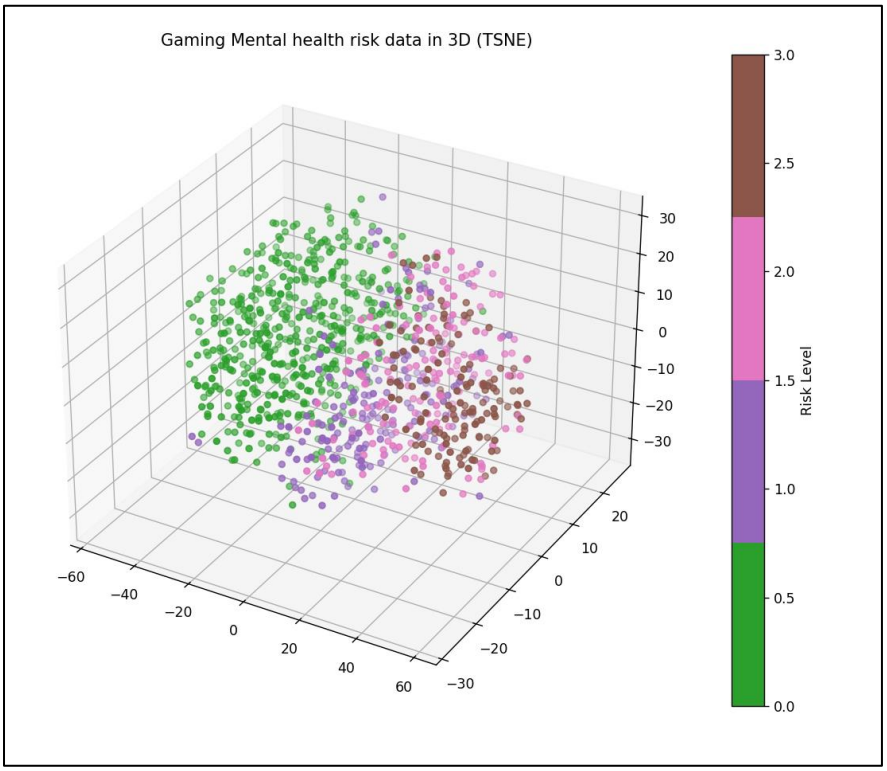

Figure 2.4: 3D representation of Gaming Mental Health Risk dataset

The 3D visualization of the dataset did not show any distinct clustering, as the data was represented in one roughly spherical shape. However, there was distinct separation in the classes when color coded by the target variable (gaming addiction risk level). The low-risk class (0), shown in green, was clearly separated from the other three classes on the left side of the plot. The moderate- (1) and high-risk (2) classes were intermixed in most views but appeared to be on two intersecting planes, which indicated some separation between them, albeit messy. The severe-risk (3) class, shown in brown, was not as clearly separated as the low-risk class, but did appear more distinct in some views and was better separated than the middle two classes. The result of t-SNE on the dataset indicates that datapoints of a class are generally most similar to points of the same class, since each class grouping is distinguishable from the others. Additionally, the placement of the class groupings indicates that the points belonging to class 3 (severe risk) are most dissimilar to the points belonging to class 0 (low risk). Both of these results were encouraging and indicate that this dataset could be viable for classification.

3. METHODOLOGY

3.1 Experimental Setup

Three models were chosen for evaluation and classification of this dataset: one unsupervised and two supervised. The unsupervised model was a k-means clustering model, which was run on the full dataset and did not use a test-train split. The selected supervised models were Logistic Regression and Multi-Layer Perceptron (MLP). Both supervised models were run on the full dataset as well as a subset identified by the correlation analysis outlined in section 2.2.1. An 80/20 train-test split was used for both supervised models. A validation split was also used for the MLP and was split off from the train section, so that the final split was 64/20/16 for train/test/val.

Multiple versions of binary and multi-class Logistic Regression models were run to compare outputs. First, in order to use a binary Logistic Regression model, two different binary masks were applied to the target. Following the results seen in Section 2.2.1, the first binary mask split the target into a “one-versus-all" mask for “low” risk versus all other risk levels, where “low” was 1 and all other risk levels were 0. The second binary mask split the possible target values in half, where “low” and “moderate” were 0 and “high” and “severe” were 1.

Second, a multi-class Logistic Regression model was run using a “one-versus-all" technique. For this model, four one-vs-all binary masks were applied to the target to produce four binary target vectors. A second version of the multi-class model was run on the full dataset with only three classes, where the middle two classes (“moderate” and “high”) were combined.

All data was standardized using z-score standardization before running any of the models. Min-max normalization was also used for the MLP.

3.2 Unsupervised Models

3.2.1 K-means

K-means clustering was applied to the dataset as an unsupervised machine learning method. The initial clusters were visualized using PCA (see Figure 3.1). To determine the optimal hyperparameter, k, the elbow method was applied to the data and the results can be seen in Figure 3.2 which shows the inertia values for k=1 to k=24. By visual inspection, it was clear that either k=3 or k=4 was the elbow point. Both values of k were used and then the Davies-Bouldin (DB) index and purity were calculated to determine the optimal number of clusters and to evaluate model performance.

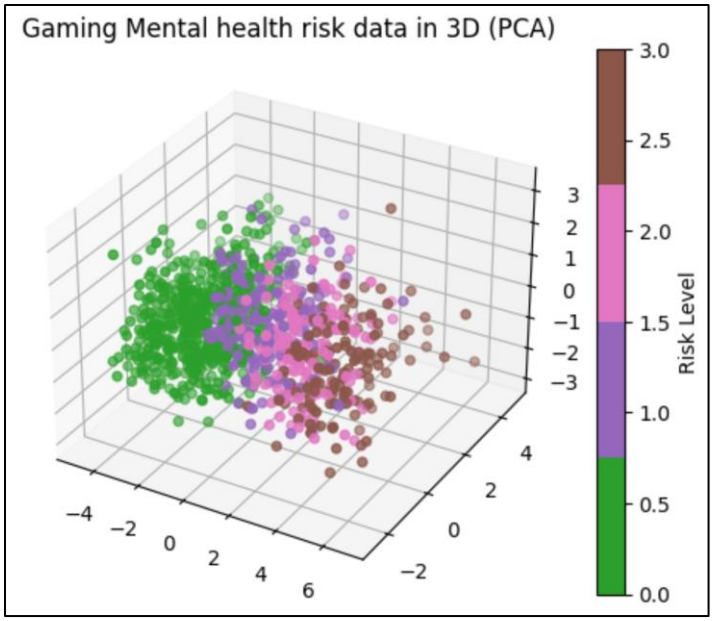

Figure 3.1: Separation of true labels using top three principal components

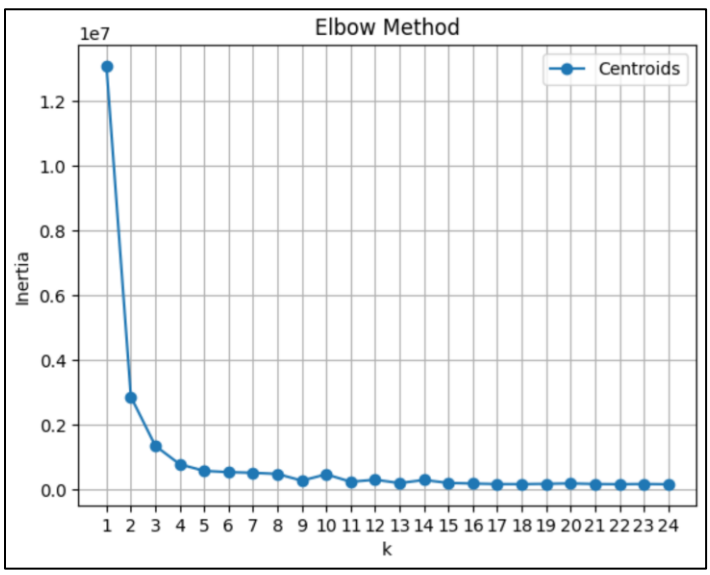

Figure 3.2: Elbow method results for determining the optimal k

The centroids were randomly initialized based on the minimum and maximum values of each feature. The resulting clusters for both k values are shown in Figures 3.3-4. The x, y, and z axis represent the top three principal components determined using PCA and purity along with the Davies-Bouldin index were used as evaluation metrics for the resulting clusters. Since this was an unsupervised method, metrics like accuracy or AUC-ROC could not be applied.

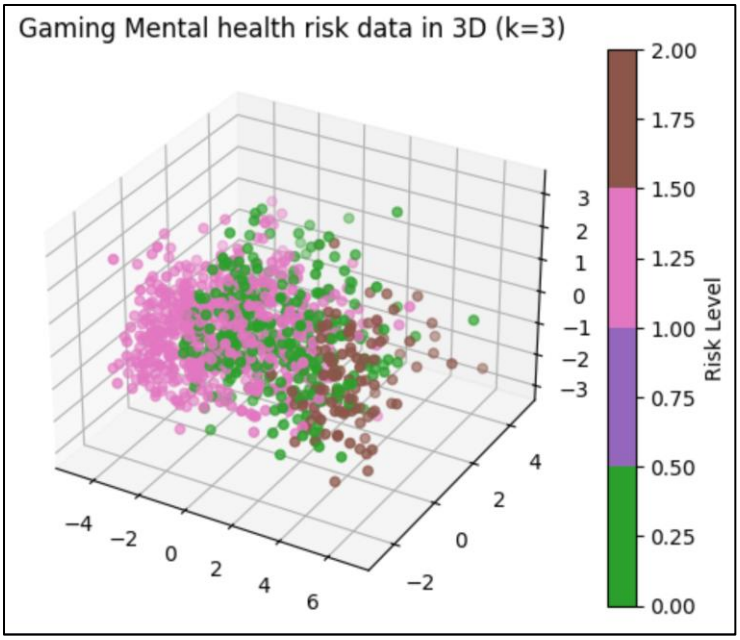

Figure 3.3: k-means clustering results for k=3

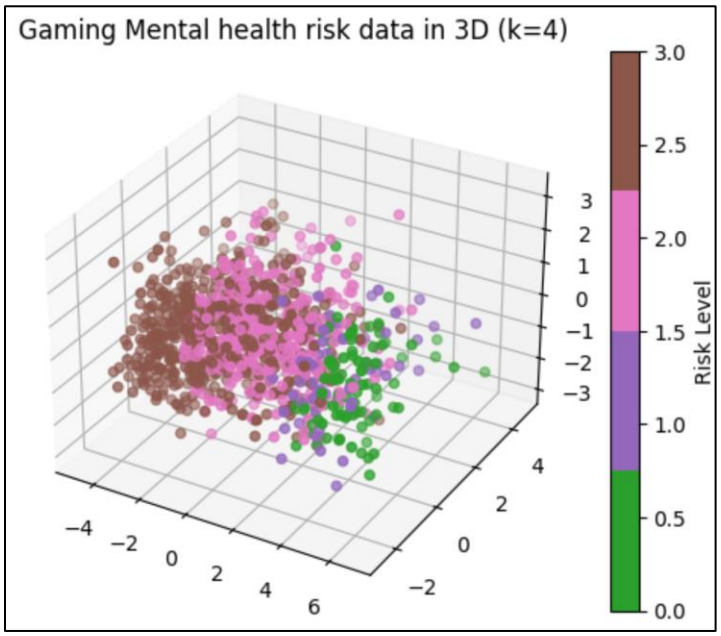

Figure 3.4: k-means clustering results for k=4

3.3 Supervised Models

3.3.1 Logistic Regression

Both binary and multi-class logistic regression models were implemented and used on the Gaming Addiction Risk dataset. No interactions were added into the dataset for any

models. The regularization control term ‘C’ was set to 10 for weaker regularization, since the number of features was very low compared to the number of features, and the dataset was not particularly noisy. The scikit-learn Logistic Regression model uses a tolerance of 0.001 for convergence, 100 maximum iterations, and the ‘lbfgs’ solver (Limited-Memory Broyden-Fletcher-Goldfarb-Shanno) by default, and these hyperparameters were left unchanged. Reported metrics for the binary logistic regression models were F1-score, accuracy, and Area Under the Curve (AUC). Accuracy was selected as a general metric, and F1 was chosen as a balance between precision and recall, since it was not clear which type of error was worse for this problem. AUC was selected to make sure the decision boundary was not negatively affecting prediction accuracy.

Since there were four possible target values for this dataset, the multi-class Logistic Regression models consisted of either three or four one-vs-all binary logistic regression models. The positive class probability vectors for each of these models were combined into a matrix, and the predicted multi-class labels were assigned based on which model produced the highest positive class probability for the given point. Metrics for multi-class logistic regression were accuracy, macro-F1, and weighted-F1. Macro-F1 was chosen to see the performance when all classes were treated equally, and weighted-F1 was chosen to see the performance when accounting for class imbalance. AUC was not reported for multi-class logistic regression since it would have had to be aggregated across all binary models instead of being reported directly on the final predictions.

3.3.2 Multi-Layer Perceptron

Two MLPs were implemented on the dataset. Both had the same underlying parameters, the only difference being the hidden layer sizes. The activation function used was ReLU and both models used the Adam optimizer. The learning rate was 0.001 and the maximum number of iterations was 1000. The ‘alpha’ parameter, which represents the strength of the L2 regularization term, was set to 0.001. The first model had one hidden layer with 64 neurons, and the second model had two hidden layers with 128 neurons and 64 neurons.

To evaluate the MLPs, four different evaluation metrics were used: accuracy, precision, recall, and weighted F1. Weighted F1 was used because of the heavy class imbalance discussed earlier in section 2.2.1. As an exploratory investigation, both models were also run on the dataset after applying normalization with two different methods: z-score and min-max normalization.

4. RESULTS

4.1 K-Means Evaluation Metrics

The metric results for k-means clustering are shown in Table 4.1. Based on the results from the elbow method, both k=3 and k=4 were tested as the optimal number of clusters. The DB index was 0.505 for k=3 and 0.532 for k=4. Since lower values indicate

better defined clusters, this implied that k=3 was optimal. Purity, which uses the true class labels in its calculations, was also calculated for the different values of k. It was expected that k=4 would have higher purity since there were four classes, but the results showed better purity with three clusters. This suggests that at least two of the classes have some overlap or are not clearly separable. For this reason, k=3 was chosen as the optimal number of clusters.

k	Davies-Bouldin Index	Purity
3	0.505	0.638
4	0.532	0.620

Table 4.1: Comparison of metrics for k-means clustering

4.2 Logistic Regression Evaluation Metrics

In total, five Logistic Regression (LR) models were run. Two of these were binary LR models, and three were multi-class LR models. The results of each model can be seen below in Tables 4.2 and 4.3.

Model	F1-Score (%)	Accuracy (%)	AUC
1. Low vs Rest	99.5	99.5	1.00
2. Low/Moderate vs High/Severe	99.1	99.5	1.00

Table 4.2: Binary Logistic Regression Metrics

Both binary LR models showed very good performance when measured by the chosen metrics. This was expected, since the low-risk and severe-risk classes appeared the most separated during EDA, and the data splits used for the binary models kept these two classes separated. The accuracy and AUC were identical for both models, and the F1 score was very similar (although very slightly lower for the second model). The results of the first model confirmed that there was enough difference between the “low” risk class and all other classes to definitively identify low-risk individuals, which was also expected since there was imbalance towards low-risk individuals in the dataset. The results of the second model indicate that when the four classes are combined into two classes (Low/High risk), there was also enough difference between the two to definitively identify individuals from either group. This is most likely because of the stronger separations in the low-risk class and severe-risk class.

Model	Weighted F1 (%)	Macro F1 (%)	Accuracy (%)
1. Full Dataset (4 classes)	89.1	83.6	89.0
2. Full Dataset (3 classes)	95.4	97.0	97.0
3. Subset (4 classes)	63.3	48.0	66.0

Table 4.3: Multi-class Logistic Regression Metrics

The multi-class LR models showed lower performance values than the binary LR models, which was expected since these models had to differentiate between the two intermediate classes (moderate-risk and high-risk), which were found to be more intermixed during EDA. The best performing multi-class model was the 3-class model,

where the intermediate classes were combined, which outperformed the 4-class model by a significant amount. The model run on the selected 6-feature subset was also a 4-class model, but it did not perform as well as the models run on the full dataset.

4.3 Multi-Layer Perceptron

Tables 4.4-6 show the metric results for both Multi-Layer Perceptron models, and Table 4.4 shows the results for the validation set with the min-max normalized data. All metrics for both models were approximately equal across all tests. The 1-Layer model with min-max scaling achieved 95% universally across all metrics, compared to the 2-Layer model's 97%.

	1-Layer	2-Layer
Accuracy	0.950	0.970
Precision	0.950	0.971
Recall	0.950	0.970
Weighted F1	0.950	0.970

Table 4.4: Min-Max normalization validation set results for both MLPs

Table 4.5 shows the results of both models using z-score normalization on the data. Similar to when min-max scaling was used, the models performed almost identically across all metrics. The 1-Layer model achieved 94% universally compared to the 2-Layer model's 96%.

	1-Layer	2-Layer
Accuracy	0.940	0.960
Precision	0.939	0.960
Recall	0.940	0.960
Weighted F1	0.939	0.960

Table 4.5: Z-score normalization validation set results for both MLPs

Table 4.6 shows the application of both models on the chosen 6-feature subset from section 2.2.1. When using only 6 features, both models were still able to achieve comparable results to the full dataset. The 1-Layer MLP achieved 93% universally, and the 2-Layer MLP achieved 95.5%.

	1 Layer	2 Layer
Accuracy	0.930	0.955
Precision	0.931	0.955
Recall	0.930	0.955
Weighted F1	0.930	0.955

Table 4.6: Feature subset validation set results

5. ANALYSIS & DISCUSSION

5.1 Comparison of results

The results of the k-means clustering model fell more under the hat of EDA for this project, since the true labels for the data were known. However, the resulting optimal number of clusters, k=3, was supported by the results found with the 3-class LR model.

The binary LR models showed very high performance, with all metrics appearing close to 100%. These results indicated a high degree of separation between the low-risk and severe-risk classes, but were most likely biased due to the large imbalance towards the low-risk class and were therefore not particularly meaningful.

The multi-class LR models showed more plausible classification results, with performance ranges between 83-97% on the full dataset. The 3-class model performed better than the 4-class model, with an accuracy value of 97% and a weighted F1 of 95.4%. This confirmed the results of the k-means clustering model and the tSNE visualization. However, the model run on the 6-feature subset performed much worse than desired, with an accuracy and weighed-F1 of 66% and 63% respectively.

The 2-layer MLP performed the best of all the models when doing multi-class classification. This model was able to achieve 96% accuracy with standard scaling on the full dataset. The 2-layer MLP also had an accuracy of 95.5% on the 6-feature subset from section 2.2.1, which supported the ideas from [1]. Also, the effect of the different normalization techniques on model performance was negligible.

5.2 Limitations

One identified limitation was the origin of the dataset. The Gaming and Mental Health dataset was available on Kaggle, but there were no identified sources listed on its data card, and it was unclear whether the dataset was real or synthetic data. The dataset also had a large number of missing values across two features and was imbalanced towards the low-risk class. The class imbalance could produce models prone to bias towards the low-risk class, and the imputation of missing values could produce results unrealistic to the rest of the dataset. Overall, the dataset had some inherent limitations that could impede results.

Secondly, some of the steps taken during the EDA and preprocessing steps of this project were done out of order. Normalization and standardization of the dataset was performed during individual model implementation, rather than once on the entire dataset during preprocessing. This likely did not have a large impact on results, since the same standardization method was used across all models, but to get the best results it should have been done as a preprocessing step.

5.3 Next Steps

The next steps on this project might include running other feature selection methods to determine a better subset to match the results found in [1]. The MLP model used for this project showed promising results, but more work could be done to find a more optimal subset of features for classification. An ablation study could be done to reduce the number of features, or a method such as forward selection or backward selection could

be used to determine the best subset of features. This subset could then be compared to the symptoms listed in [1] and the results of this project to see where they agreed and disagreed.

VI. CONCLUSIONS

The results drawn from the EDA methods used in this project indicated that the Gaming and Mental Health dataset would be viable for classification, despite a class imbalance and missing values. Spearman correlation was performed and indicated that there was meaningful correlation present in the dataset, but not enough to clearly identify any redundant features. Reduced dimensionality visualization showed clear groupings in the data, and the data ranges and statistics indicated real-world distributions. From these methods, it was determined that the dataset could be used for classification.

Results from the selected machine learning models showed encouraging performance results for classification, both for the full dataset and for a selected 6-feature subset. The multi-class Logistic Regression model used on the full dataset achieved 89% accuracy on the full dataset, and the 2-layer multi-layer perceptron achieved 96% accuracy on the full dataset. The 2-layer multi-layer perceptron achieved 95% accuracy on the 6-feature subset, which was almost as high as the full dataset. The 6-feature subset was determined from both the Spearman correlation table results and [1], and prediction results confirmed that the symptoms listed in [1] were meaningful for prediction of gaming addiction risk level based on this dataset.

VII. AUTHOR CONTRIBUTIONS

Rachael Ballentine: Conceptualization, Data Curation, Methodology, Software, Formal Analysis, Validation, Writing – Original Draft
Diana Johnson: Conceptualization, Methodology, Software, Formal Analysis, Validation, Writing – Original Draft, Writing – Review & Editing

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